AD156A Datasheet

Zhuhai Jieli Technology Co.,LTD

Version：V1.4

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ZHUHAI JIELI TECHNOLOGY CO.,LTD

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AD156A Features

CPU Core

- 32-bit CPU,Built-in ICACH, can be connected to Flash for expansion of code
- The main frequency is up to 120MHz

Memory

- On-chip SRAM 20kbyte
- Built-in Flash memory

Clock Source

- RC clock frequency about 16MHz
- LRC clock frequency about 200KHz,Accuracy ±1% at -40℃ to +85℃

Digital I/O

- Up to 28 programmable digital I/O pins
- General the IO supports pull-up(10k),pull-down(60k), strong,weak output,input and high impedance
- Up to 12 external interrupt/wake-up source(low power available,can be multiplexed to any I/O, with hardware filter)
- Input channel and Output channel, provide arbitrary IO input and output options for some modules

Digital peripherals

- Two UART Controllers(UART0/1) supports DMA and Flow Control
- Two SPI Controllers with DMA(SPI0/1) support master mode and slave mode,SPI0 support 4bit,SPI1support 2bit
- Built-in Flash for code
- One SD host controller
- Three 32-bit Asynchronous Divider Timers
- One IIC Controller
- Four channel PWM output
- Infrared remote control decoder
- Watchdog

Analog Peripherals

- 0.5 watt Class-D audio amplifier output
- 14 channel 10-bit high precision ADC
- Low voltage protection
- Power on reset

PMU

- Soft off current: <2uA
- Power down current: <30uA
- Music mode: <6mA
- LVD range(3bit):1.8V~2.5V, step100mV
- VBAT range : 2.0V~5.5V
- VDDIO range : 2.0V~3.4V

Temperature

- Operating temperature: -40℃ to +85℃
- Storage temperature: -65℃ to +150℃

Package

- QFN32(4mm*4mm)

Application

- Sound Toy
- Audio player
- Universal Microcontroller

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1 Block Diagram

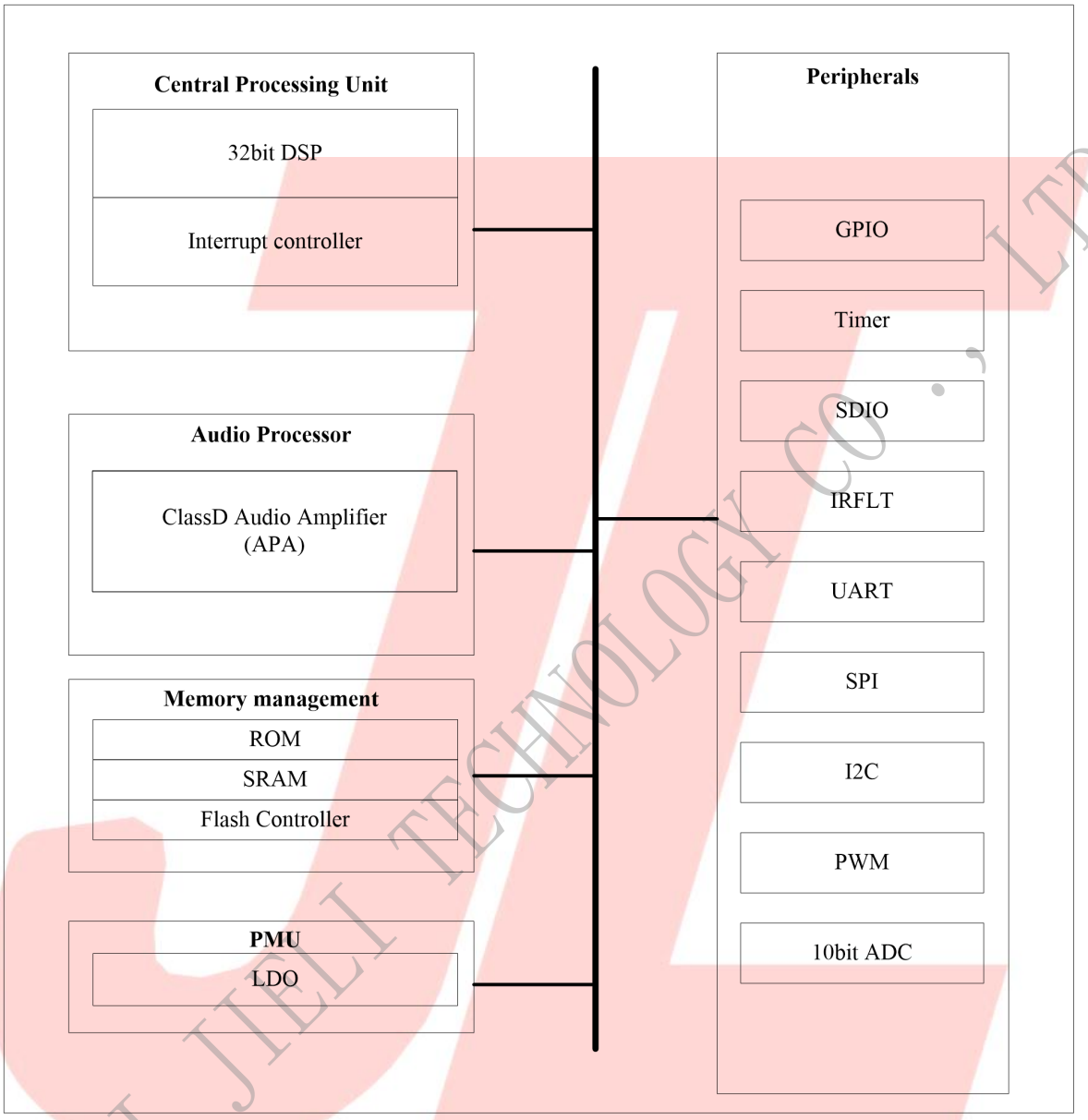

Figure 1-1 AD156A Block Diagram

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2 Pin Definition

2.1 Pin Assignment

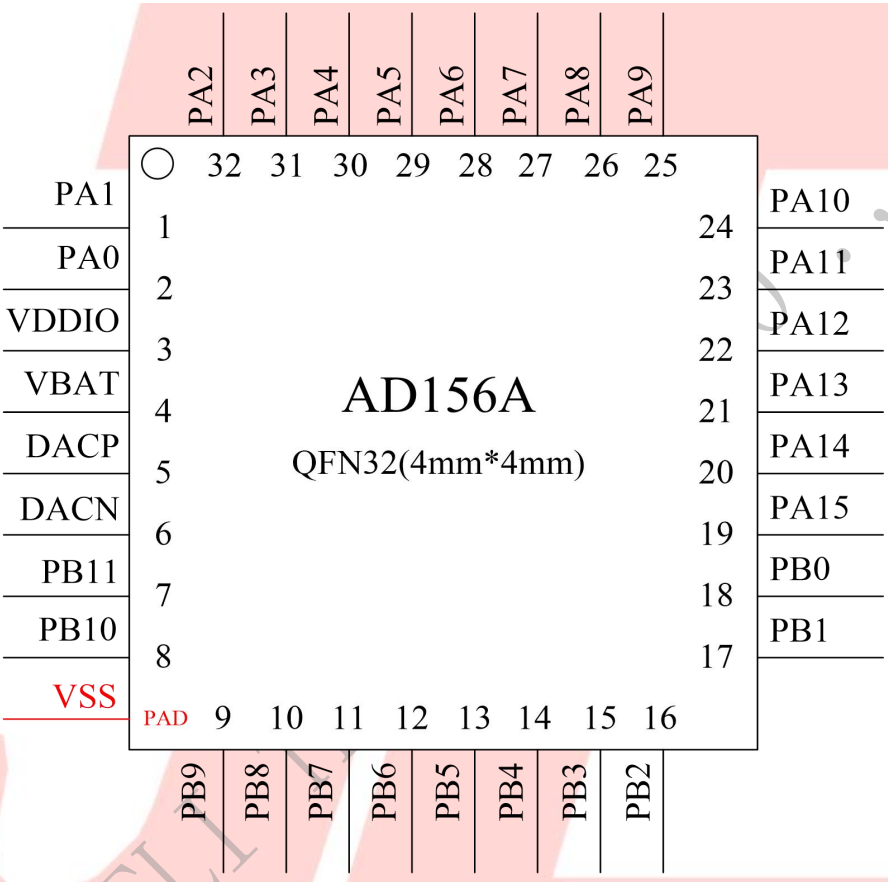

Figure 2-1 AD156A_QFN32 Package Diagram

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2.2 Pin Description

Table 2-1 AD156A_QFN32 Pin Description

PIN NO.	Name	Type	Drive (mA)	Function	Description
1	PA1	I/O	8/64	GPIO	ADC1:ADC Input Channel 1; SPI0CLKB:SPI0 Clock(B); SD0CLKA:SD0 Clock(A); UART0RXB:Uart0 Data In(B); I2C_SCL(B); CAP2:Timer2 Capture;
2	PA0	I/O	8/64	GPIO (pull up)	Long Press Reset; ADC0:ADC Input Channel 0; UART0TXB:Uart0 Data Out(B);
3	VDDIO	P	/		Digital Power; (Internal linear regulator output)
4	VBAT	P	/		Battery Power Supply;
5	DACP	O	/		Class-D APA Positive Output;
6	DACN	O	/		Class-D APA Negative Output;
7	PB11	I/O	8	GPIO (High Voltage Resistance)	OSCIB:Crystal Oscillator Input(B);
8	PB10	I/O	8	GPIO (pull up) (High Voltage Resistance)	MCLR(0 effective);
9	PB9	I/O	8	GPIO (High Voltage Resistance)	SPI1DOD:SPI1 Data Out(D); UART1TRXB:Uart1 Data In/Out(B); I2C_SDA(D); CAP1:Timer1 Capture;
10	PB8	I/O	8	GPIO (High Voltage Resistance)	SPI1CLKD:SPI1 Clock(D); I2C_SCL(D); OSCIA:Crystal Oscillator Input(A);
11	PB7	I/O	8/64	GPIO	SPI1DID:SPI1 Data In(D);
12	PB6	I/O	8/64	GPIO	SD0DATC:SD0 Data(C);
13	PB5	I/O	8/64	GPIO	ADC13:ADC Input Channel 13; SD0CMDC:SD0 Command(C);
14	PB4	I/O	8/64	GPIO	ADC12:ADC Input Channel 12; SD0CLKC:SD0 Clock(C);
15	PB3	I/O	8/64	GPIO	TDM_MCLK;

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16	PB2	I/O	8/64	GPIO	SPI1DIA:SPI1 Data In(A); SD0DATB:SD0 Data(B); TDM_DAT;
17	PB1	I/O	8/64	GPIO (pull down)	ADC11:ADC Input Channel 11; SPI1DOA:SPI1 Data Out(A); SD0CMDB:SD0 Command(B); I2C_SDA(A); TDM_SYN;
18	PB0	I/O	8/64	GPIO (pull down)	ADC10:ADC Input Channel 10; SPI1CLKA:SPI1 Clock(A); SD0CLKB:SD0 Clock(B); I2C_SCL(A); TDM_CLK;
19	PA15	I/O	8/64	GPIO	ADC9:ADC Input Channel 9; SPI1DOB:SPI1 Data Out(B); MCAP3:Motor Timer3 Capture;
20	PA14	I/O	8/64	GPIO	ADC8:ADC Input Channel 8; SPI1CLKB:SPI1 Clock(B); CAP0:Timer0 Capture; MCAP2:Motor Timer2 Capture;
21	PA13	I/O	8/64	GPIO	SPI1DIB:SPI1 Data In(B); TMR1:Timer1 Clock In; MCAP1:Motor Timer1 Capture;
22	PA12	I/O	8/64	GPIO	MPWM3:PWM Channel3 Output;
23	PA11	I/O	8/64	GPIO	TMR0:Timer0 Clock In; MPWM2:PWM Channel2 Output;
24	PA10	I/O	8/64	GPIO	
25	PA9	I/O	8/64	GPIO	
26	PA8	I/O	8/64	GPIO	SPI1DIC:SPI1 Data In(C); SD0DATD:SD0 Data(D);
27	PA7	I/O	8/64	GPIO	ADC7:ADC Input Channel 7; SPI1DOC:SPI1 Data Out(C); SD0CMDD:SD0 Command(D); UART0RXA:Uart0 Data In(A); I2C_SDA(C); MPWM1:PWM Channel1 Output;

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28	PA6	I/O	8/64	GPIO	ADC6:ADC Input Channel 6; SPI1CLKC:SPI1 Clock(C); SD0CLKD:SD0 Clock(D); UART0TXA:Uart0 Data Out(A); I2C_SCL(C); TMR2:Timer2 Clock In; MPWM0:PWM Channel0 Output;
29	PA5	I/O	8/64	GPIO	ADC5:ADC Input Channel 5; SPI0DAT3:SPI0 Data 3 UART1RXA:Uart1 Data In(A);
30	PA4	I/O	8/64	GPIO	ADC4:ADC Input Channel 4; SPI0DAT2:SPI0 Data 2; UART1TXA:Uart1 Data Out(A); LVD:Low Voltage Detect;
31	PA3	I/O	8/64	GPIO	ADC3:ADC Input Channel 3; SPI0DIB(1):SPI0 Data1 In(B); SD0DATA:SD0 Data(A); CLKOUT; PWM2(B); MCAP0:Motor Timer0 Capture;
32	PA2	I/O	8/64	GPIO	ADC2:ADC Input Channel 2; SPI0DOB(0):SPI0 Data0 Out(B); SD0CMDA:SD0 Command(A); I2C_SDA(B); PWM2(A);
PAD	VSS	G			Ground;

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3 Electrical Characteristics

3.1 Absolute Maximum Ratings

Table 3-1

Symbol	Parameter	Min	Max	Unit
Tamb	Ambient Temperature	-40	+85	°C
Tstg	Storage temperature	-65	+150	°C
VBAT	Supply Voltage	-0.3	5.5	V
V_{VDDIO}	Voltage applied at VDDIO	-0.3	3.6	V
V_{GPIO}	Voltage applied to GPIO	-0.3	VDDIO+0.3	V
V_{HVIO}	Voltage applied to High Voltage Resistant IO	-0.3	+5.5	V

Note : The chip can be damaged by any stress in excess of the absolute maximum ratings listed below

3.2 PMU Characteristics

Table 3-2

Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
VBAT	Voltage Input	2.0	3.7	5.5	V	_
V_{VDDIO}	Voltage output	2.0	3.0	3.4	V	VBAT = 3.7V, 100mA loading
I_{VDDIO}	Loading current	_	_	100	mA	VBAT=3.7V

3.3 IO Input/Output Electrical Logical Characteristics

Table 3-3

GPIO input characteristics						
Symbol	**Parameter**	**Min**	**Typ**	**Max**	**Unit**	**Test Conditions**
V_{IL}	Low-Level Input Voltage	-0.3	_	0.3* VDDIO	V	VDDIO = 3.0V
V_{IH}	High-Level Input Voltage	0.7* VDDIO	_	VDDIO+0.3	V	VDDIO = 3.0V
High Voltage Resistant IO input characteristics						
Symbol	**Parameter**	**Min**	**Typ**	**Max**	**Unit**	**Test Conditions**
V_{IL}	Low-Level Input Voltage	-0.3	_	0.3* VDDIO	V	VDDIO = 3.0V
V_{IH}	High-Level Input Voltage	0.7* VDDIO	_	+5V	V	VDDIO = 3.0V
GPIO & High Voltage Resistant IO output characteristics						
Symbol	**Parameter**	**Min**	**Typ**	**Max**	**Unit**	**Test Conditions**

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V_{OL}	Low-Level Output Voltage	–	–	0.1* VDDIO	V	VDDIO = 3.0V
V_{OH}	High-Level Output Voltage	0.9* VDDIO	–	–	V	VDDIO = 3.0V

3.4 Internal Resistor Characteristics

Table 3-4

Port	General Output	High Drive	Internal Pull-Up Resistor	Internal Pull-Down Resistor	Comment
PA0~PA15 PB0~PB7	8mA	64mA	10K	60K	1、PA0&PB10 default pull up 2、PB0 & PB1 default pull down 3、internal pull-up/pull-down resistance | accuracy ±20%
PB8~PB11	8mA	–	10K	60K	

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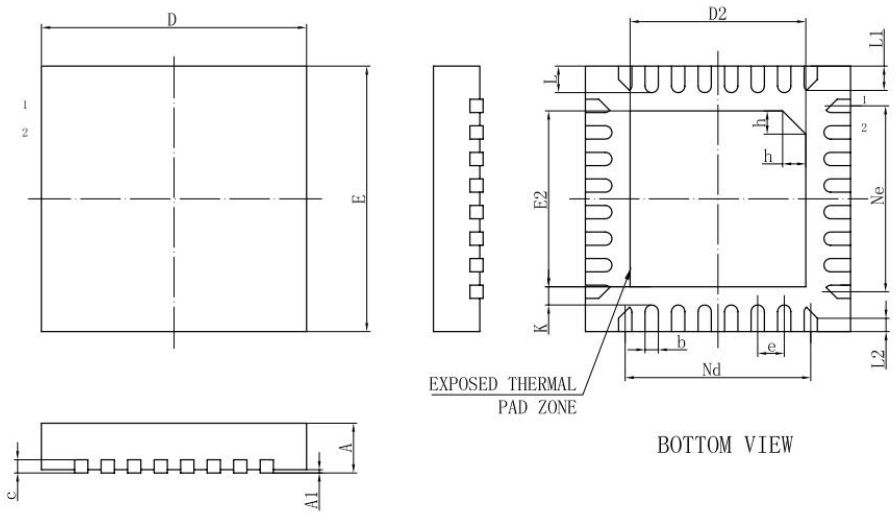
4 Package Information

4.1 QFN32



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.70	0.75	0.80
A1	0	0.02	0.05
b	0.15	0.20	0.25
c	0.18	0.20	0.25
D	3.90	4.00	4.10
D2	2.60	2.65	2.70
e	0.40BSC		
Nd	2.80BSC		
E	3.90	4.00	4.10
E2	2.60	2.65	2.70
Ne	2.80BSC		
K	0.20	–	–
L	0.35	0.40	0.45
L1	0.30	0.35	0.40
L2	0.15	0.20	0.25
h	0.30	0.35	0.40
L/F载体尺寸 (MIL)	112*112		

Figure 4-1. AD156A_QFN32 Package

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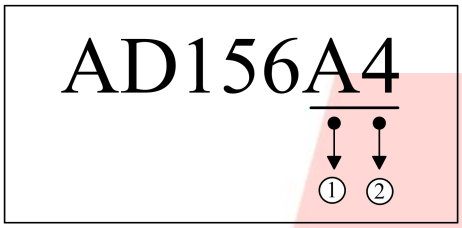
5 Package Type Specification



①Represents different packages

②Represents different memory sizes

0: No memory

2: 2Mbit Flash

4: 4Mbit Flash

8: 8Mbit Flash

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6 Storage Condition

6.1 Moisture Sensitivity Level

AD156A is qualified to moisture sensitivity level MSL3 in accordance with JEDEC J-STD-033.

6.2 Storage Alert

1. Calculated shelf life in sealed bag 12 months at＜40°C and 90% relative humidity (RH).
2. Peak package body temperature≤260°C.
3. After bag is opened,devices that will be subjected to reflow solder or other high temperature process must be mounted within 168 hours of factory conditions≤30°C/60%RH or stored per J-STD-033.
4. Devices require bake before mounting if humidity indicator card reads＞10% for level 2a-5a devices or＞60% for level 2 devices when read at 23±5°C,or 3a or 3b are not met.

Please refer to IPC/JEDEC J-STD-033 for baking procedure if necessary.

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7 Revision History

Date	Revision	Description
2021.03.03	V1.0	Initial Release
2021.05.08	V1.1	parameter modification
2021.04.27	V1.2	LRC precision description
2023.03.21	V1.3	Modify the Features.
2024.11.20	V1.4	Add Block Diagram Add Storage Condition Update Features modification.

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